

SHALINSKOYE, Russian Federation

WMO: 295780

Lat: 55.7173N

Lon: 93.7461E

Elev: 399

StdP: 96.62

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.1	-32.8	-38.7	0.1	-36.1	-35.4	0.1	-32.8	11.9	-8.0	9.6	-7.4	1.1	240	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	28.4	17.7	26.7	17.3	24.9	16.4	19.6	25.3	18.7	23.8	17.8	22.8	2.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	13.3	21.9	16.7	12.5	21.1	15.8	11.8	20.2	58.0	25.1	54.9	23.7	52.0	22.8	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.0	7.0	-38.8	31.6	4.1	1.5	-41.8	32.6	-44.2	33.5	-46.5	34.3	-49.4	35.4		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-16.0	-13.3	-4.0	3.2	8.2	16.0	17.7	15.5	8.2	1.0	-9.1	-12.4
	DBStd	13.15	9.14	8.10	6.39	4.67	4.94	4.08	2.53	3.39	4.30	5.55	7.93	8.65
	HDD10.0	3837	806	653	434	209	96	6	0	2	85	279	573	694
	HDD18.3	6259	1064	886	693	453	316	90	41	99	306	536	823	952
	CDD10.0	674	0	0	0	6	40	184	238	174	30	2	0	0
	CDD18.3	53	0	0	0	0	1	19	21	12	0	0	0	0
	CDH23.3	721	0	0	0	2	31	333	215	131	10	0	0	0
	CDH26.7	140	0	0	0	0	4	73	34	28	0	0	0	0
Wind	WSAvg	2.9	3.0	2.9	3.2	3.5	3.5	2.5	2.2	2.2	2.5	3.0	3.2	3.3
Precipitation	PrecAvg	510	15	14	19	33	47	65	73	88	53	42	34	27
	PrecMax	644	27	30	47	59	93	109	146	175	120	79	59	50
	PrecMin	398	5	5	7	9	17	16	9	49	8	12	18	5
	PrecStd	62	6	7	10	14	20	21	31	26	22	16	12	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.5	4.1	12.3	21.5	26.9	30.8	30.3	29.6	25.0	18.9	9.7	2.8
		MCWB	-0.9	-0.2	4.8	9.0	13.7	18.4	19.8	19.1	14.7	9.5	5.9	-0.4
	2%	DB	-0.2	1.8	8.6	17.0	23.8	28.7	27.7	26.6	21.5	13.9	5.0	1.2
		MCWB	-2.6	-1.1	3.2	7.7	12.9	17.6	18.1	18.4	14.1	7.9	1.7	-1.3
	5%	DB	-2.0	-0.3	6.3	13.9	20.9	27.1	25.9	24.2	18.6	11.5	3.0	-0.4
		MCWB	-4.3	-3.1	1.8	6.8	11.2	16.9	17.9	17.4	12.9	6.6	0.0	-2.4
	10%	DB	-3.9	-2.1	4.5	11.8	18.1	25.1	24.1	22.0	15.9	9.0	0.9	-2.1
		MCWB	-6.0	-4.3	0.6	5.4	10.1	16.1	17.1	16.2	11.1	5.3	-1.1	-3.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.6	0.7	5.2	10.6	14.7	20.3	20.7	20.7	15.4	10.4	6.0	0.2
		MCDB	1.1	3.3	10.4	17.9	24.8	27.9	27.0	26.6	22.3	16.1	10.5	1.9
	2%	WB	-2.6	-1.3	3.6	8.9	13.5	19.1	19.7	19.3	14.5	8.7	2.2	-1.0
		MCDB	-0.5	1.3	8.4	15.1	21.2	25.7	25.2	23.9	20.0	13.3	4.0	0.9
	5%	WB	-4.2	-2.7	2.1	7.5	12.3	18.1	18.8	18.2	13.5	7.1	0.6	-2.2
		MCDB	-1.9	-0.5	5.9	13.5	19.0	24.2	23.6	22.7	18.2	10.7	2.9	-0.4
	10%	WB	-5.9	-4.3	0.9	5.6	10.9	17.2	17.9	17.0	12.0	5.6	-1.1	-3.8
		MCDB	-3.9	-2.3	4.1	11.4	17.0	23.2	22.5	21.5	15.1	8.6	0.7	-2.0
Mean Daily Temperature Range	5% DB	MDBR	7.9	9.5	9.7	10.4	11.3	12.8	10.2	10.5	9.8	7.7	7.0	7.0
		MCDBR	7.1	10.1	11.2	13.6	17.4	16.6	14.3	14.4	15.6	13.3	7.3	7.3
		MCWBR	5.6	7.8	7.2	7.4	8.4	6.5	6.0	6.5	8.6	7.9	5.2	6.1
	5% WB	MCDBR	6.8	8.9	10.4	12.6	14.1	13.7	11.4	11.8	14.4	12.5	7.2	7.1
		MCWBR	5.6	7.2	6.9	7.3	7.5	6.0	5.4	5.8	8.1	7.8	5.6	6.0
Clear-Sky Solar Irradiance	taub	0.274	0.308	0.351	0.387	0.406	0.399	0.419	0.403	0.361	0.341	0.308	0.276	
	taud	2.035	2.017	1.981	2.134	2.204	2.317	2.283	2.293	2.364	2.261	2.076	1.981	
	Ebn at Noon	605	730	789	826	834	846	818	806	792	698	550	485	
	Edh at Noon	66	99	132	132	132	119	121	113	91	78	63	55	
All-Sky Solar Radiation	RadAvg	0.83	1.83	3.19	4.29	4.94	5.93	5.27	4.26	2.77	1.50	0.76	0.50	
	RadStd	0.09	0.11	0.25	0.27	0.53	0.49	0.37	0.38	0.35	0.12	0.10	0.06	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page